



Studio IV: Co-Living

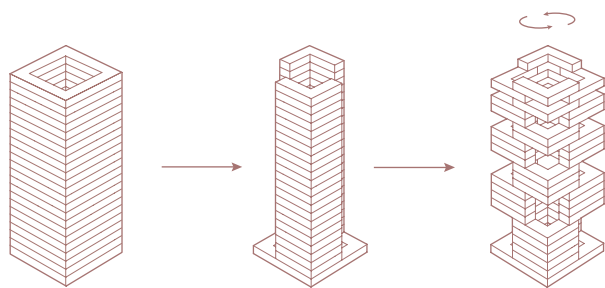
A New Grammar for Collective Housing

Columbia University - GSAPP

Selected for Summer '25 Summit

This high-rise residential concept is initially inspired by Steven Holl's Alphabet City. Here, the 'O' typology is used to extrude a block in Manhattan, allowing for maximum efficiency of surface area within the form. A series of sheared 'L' shapes are then introduced within this 'O,' each rotated around the building's central axis. This process creates dynamic shared spaces designed to foster interaction among the communities inhabiting this new typology.

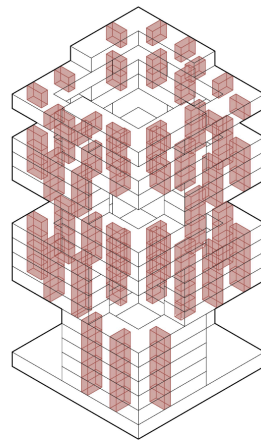
Scales of Sharing



'O' Typology

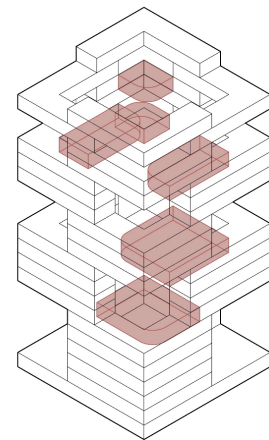
'L' Typology

'L' + 'O'



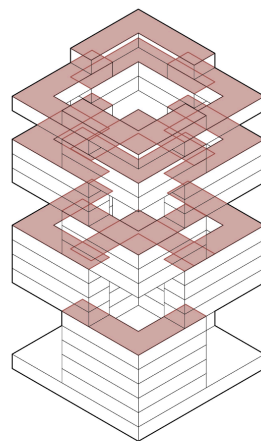
one bathroom / two units

bathrooms



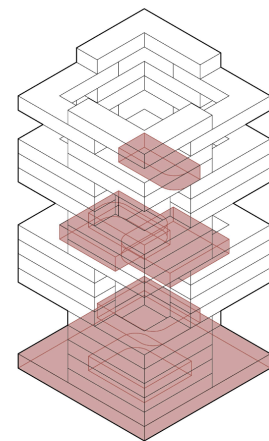
one kitchen / one cluster

kitchens dining rooms



two terraces / one cluster

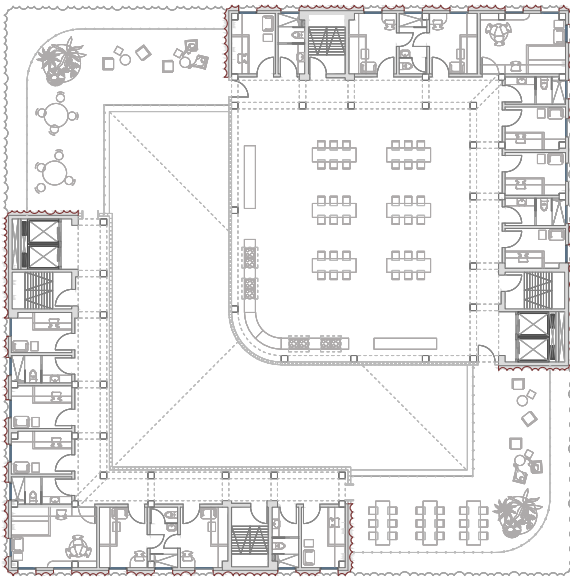
lounges terraces gardens



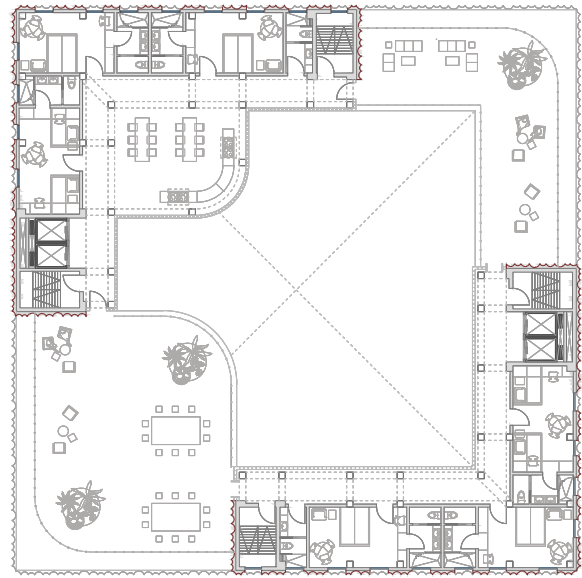
nine luxuries / whole building

library co-working makerspace
retail laundry storage cafe sports playground

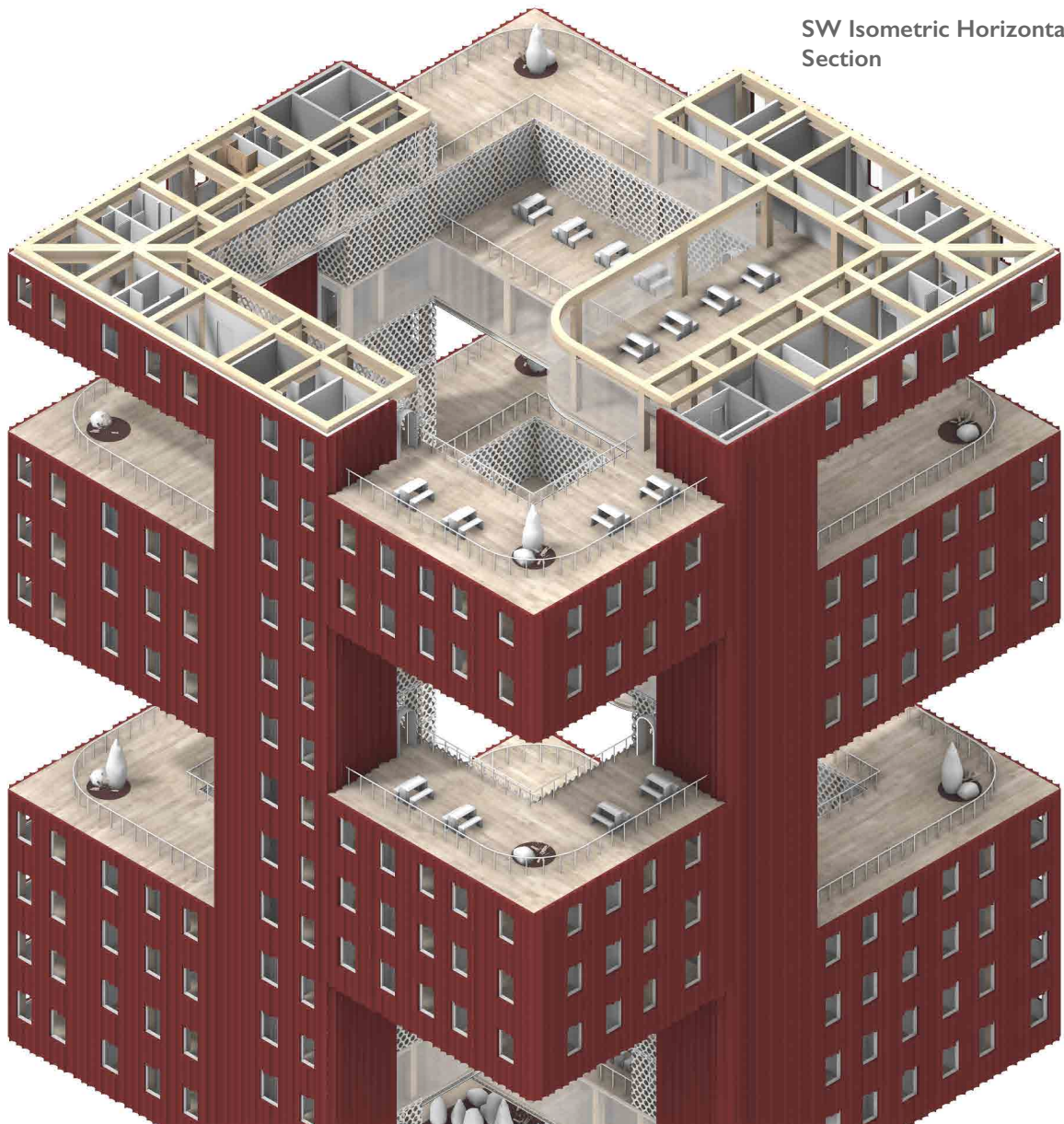




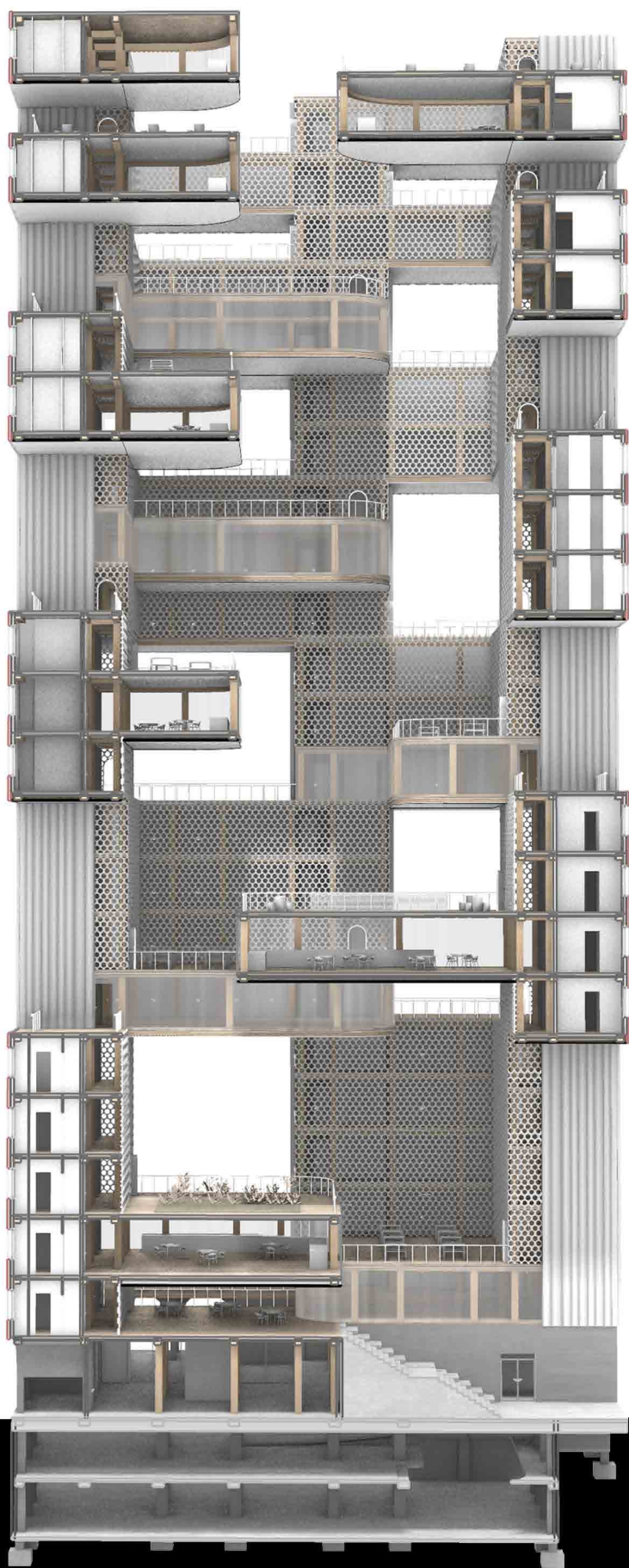
7th Floor - 14 units



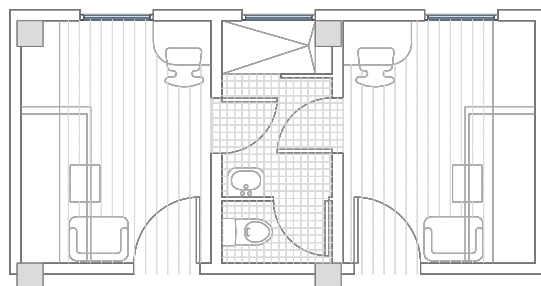
23rd Floor - 6 units



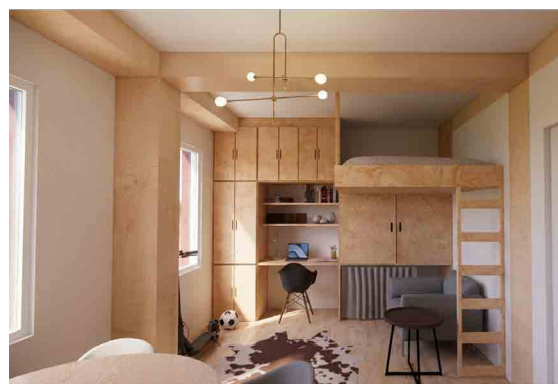
SW Isometric Horizontal
Section



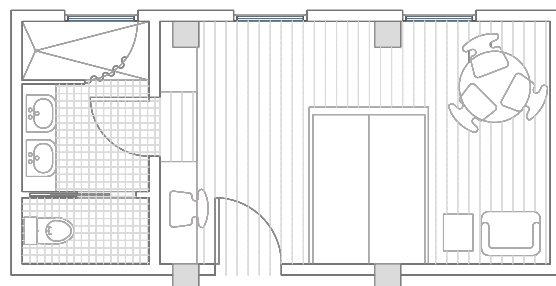
A series of distinct room typologies are distributed vertically throughout the tower, with shared programs concentrated near the base to accommodate individuals seeking single rooms. As the floors ascend, the level of privacy increases—bathrooms become private, and the living spaces expand slightly to house families and individuals who wish to live together.



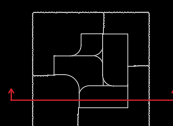
Room Type 1



View Inside Room



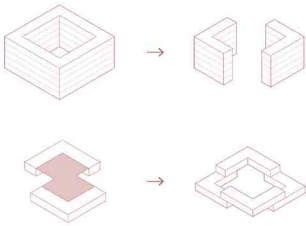
Room Type 2



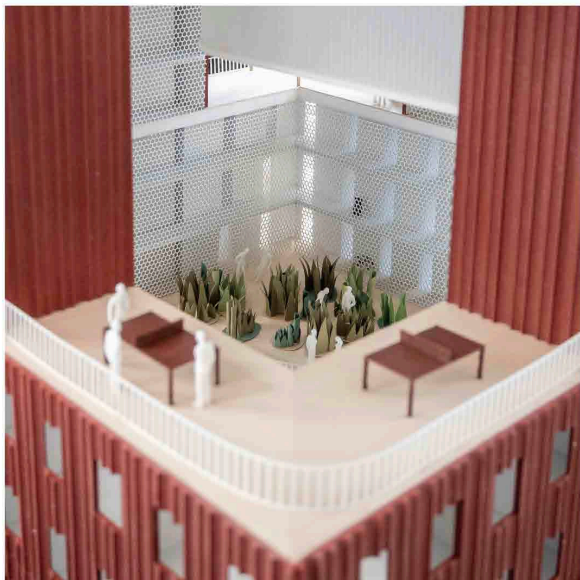
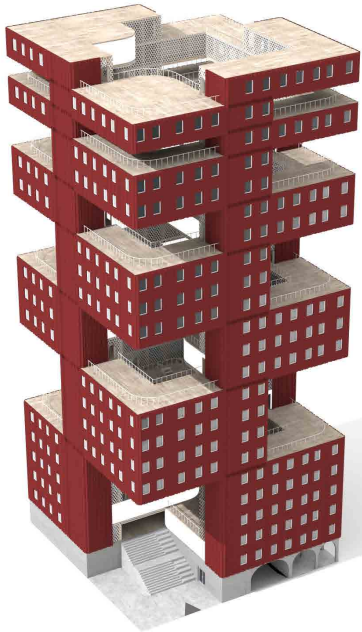
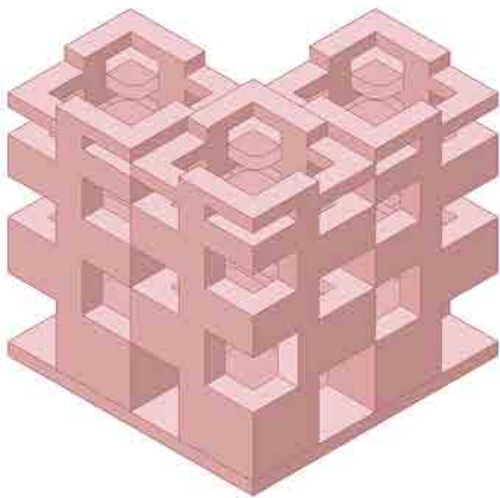
West Side Section

0 2m 10m

RECIPROCITY



The system functions best when reciprocated. For this reason, the design enables congruent models to adjoin, causing interior spatial structures to expand, thereby forming a more cohesive urban grid.



Studio V: Impermanent Matter

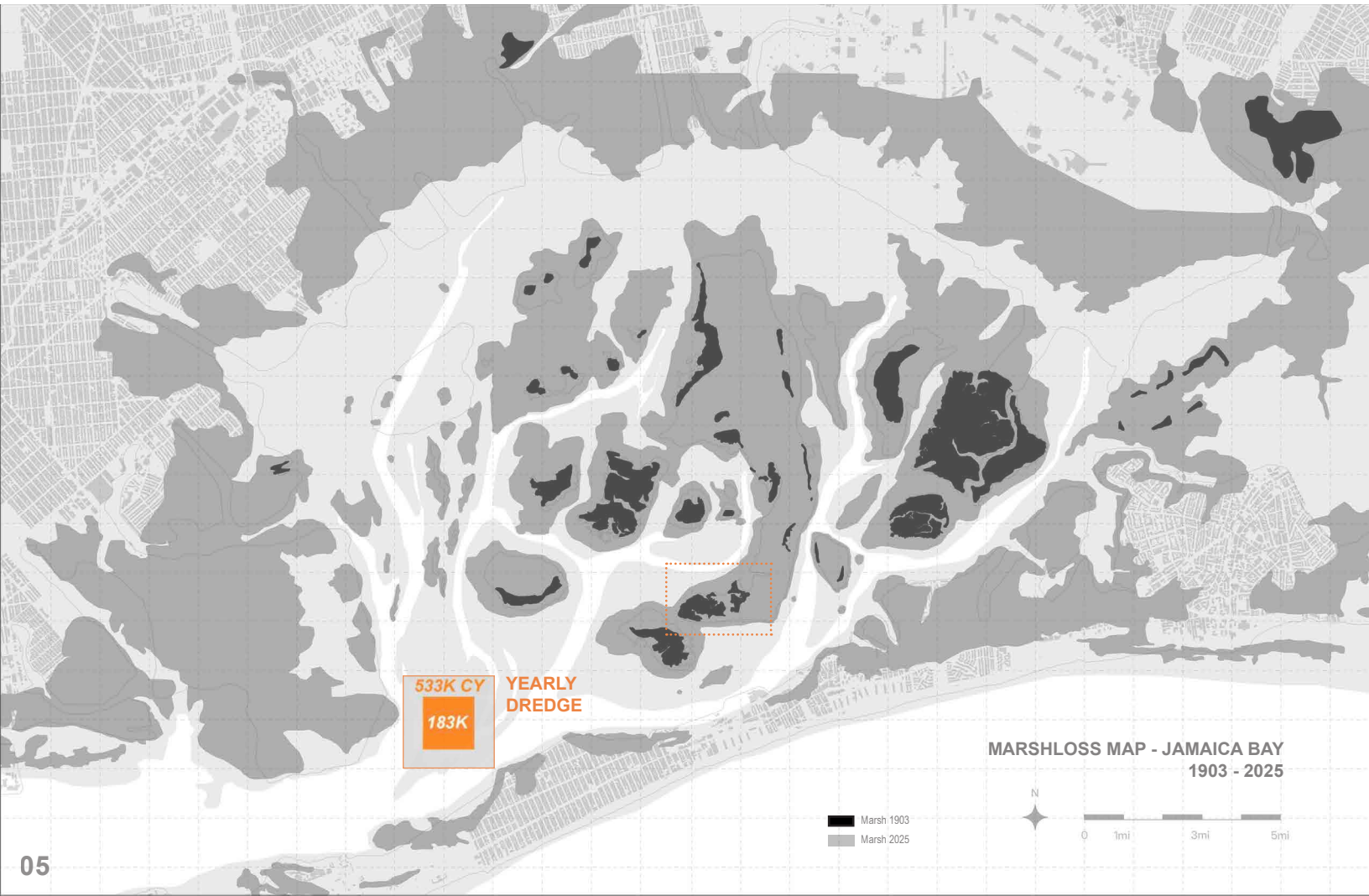
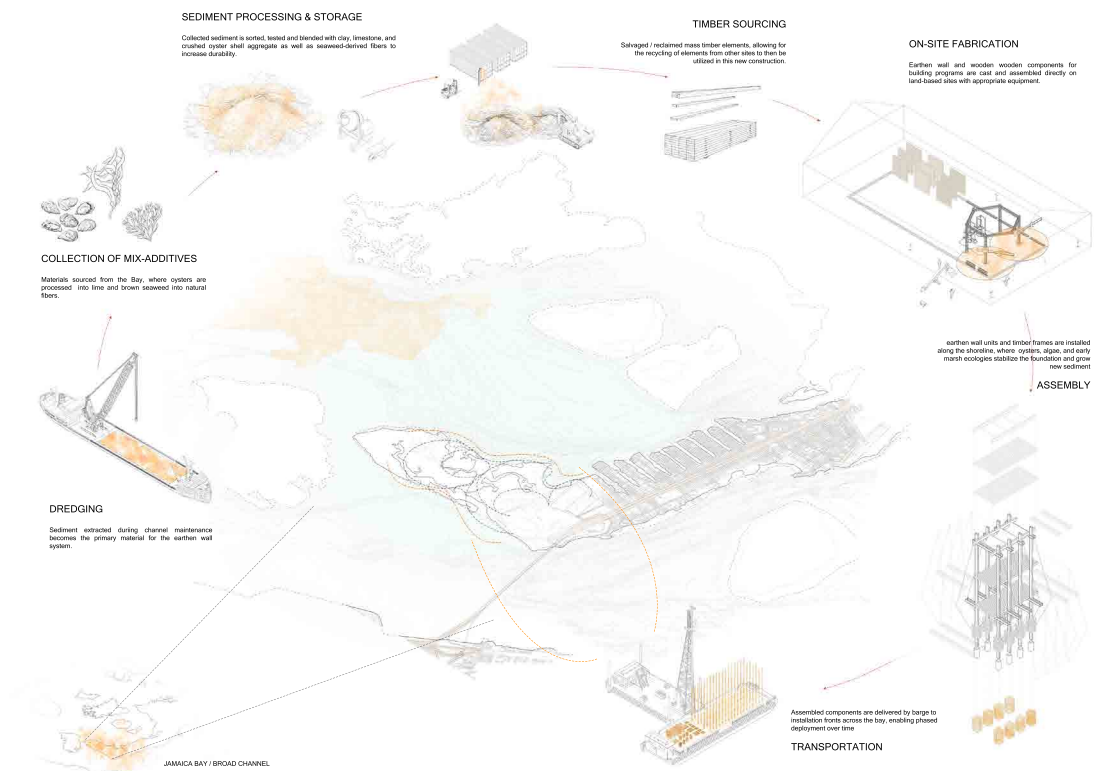
The Living Shoreline

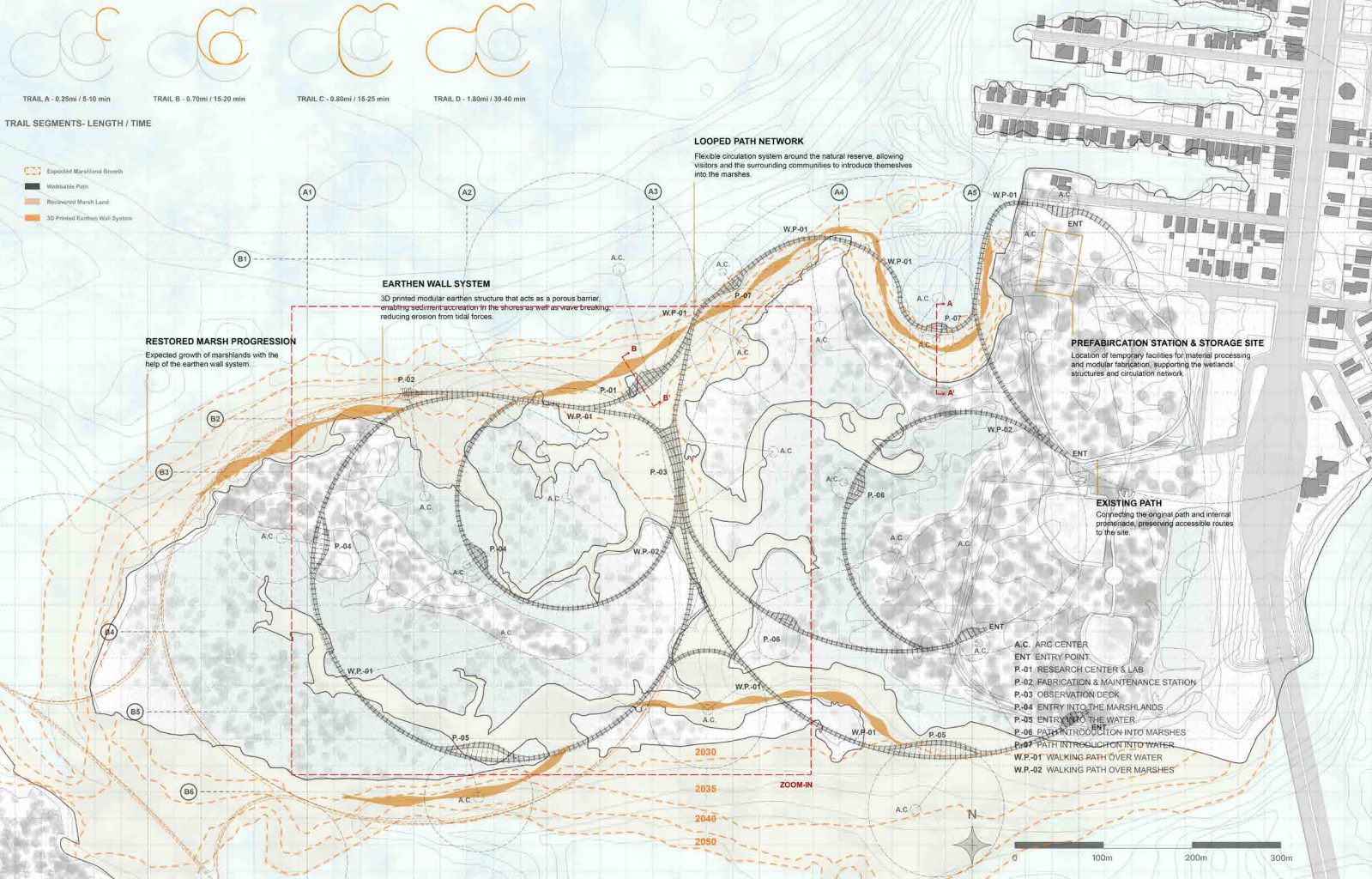
Columbia University - GSAPP
Selected for Fall '25 Summit

The project comprises a comprehensive analysis of Jamiaca Bay, an area whose ecosystem has continually been degrading for over 100 years. Most notably, salt marsh extents have been reduced to 10% of their original footprint, no longer being able to sustain the wildlife and the ecosystem that the once harbored.

The project aims to revitalize the ecosystem through a series of impermanent structures and materials that enhance and protect salt marshes.

System Sequence Scheme

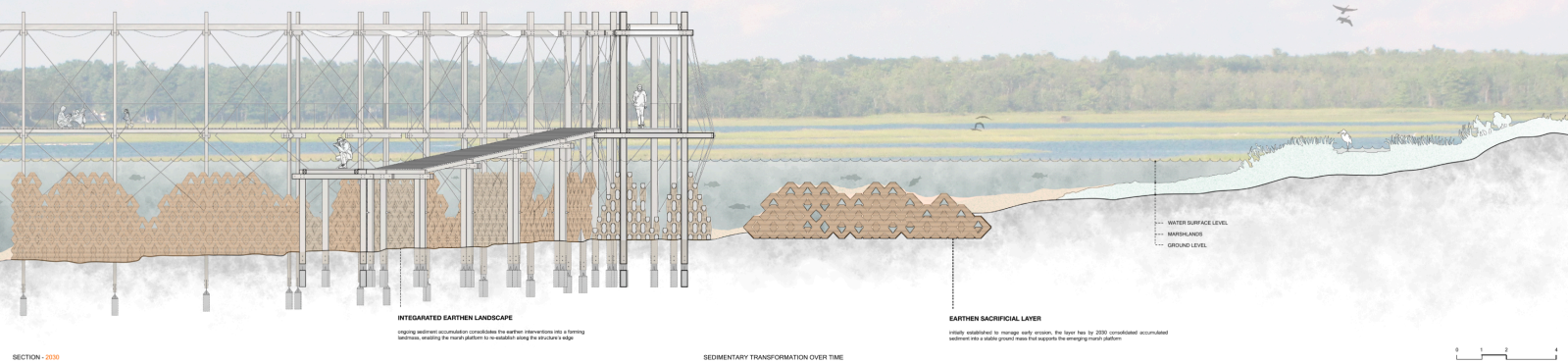


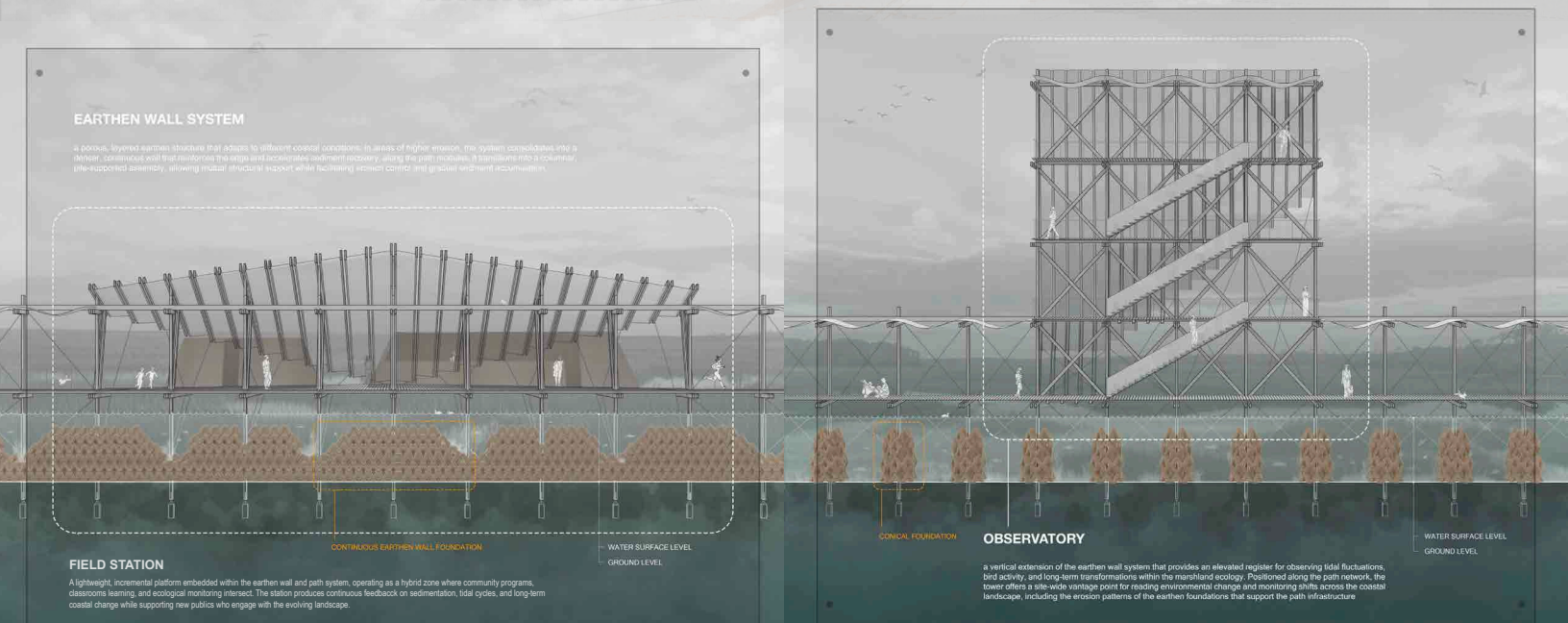
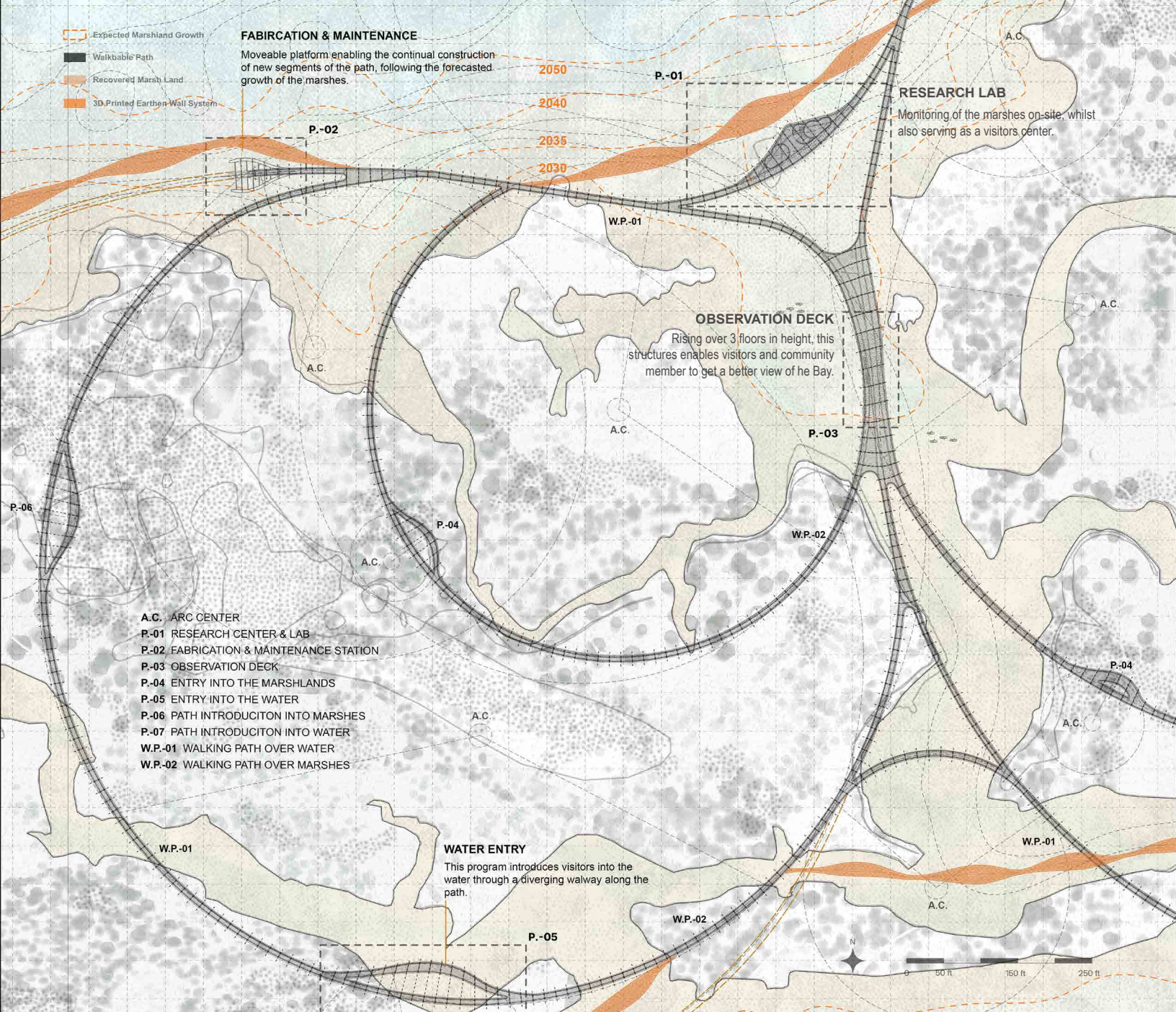


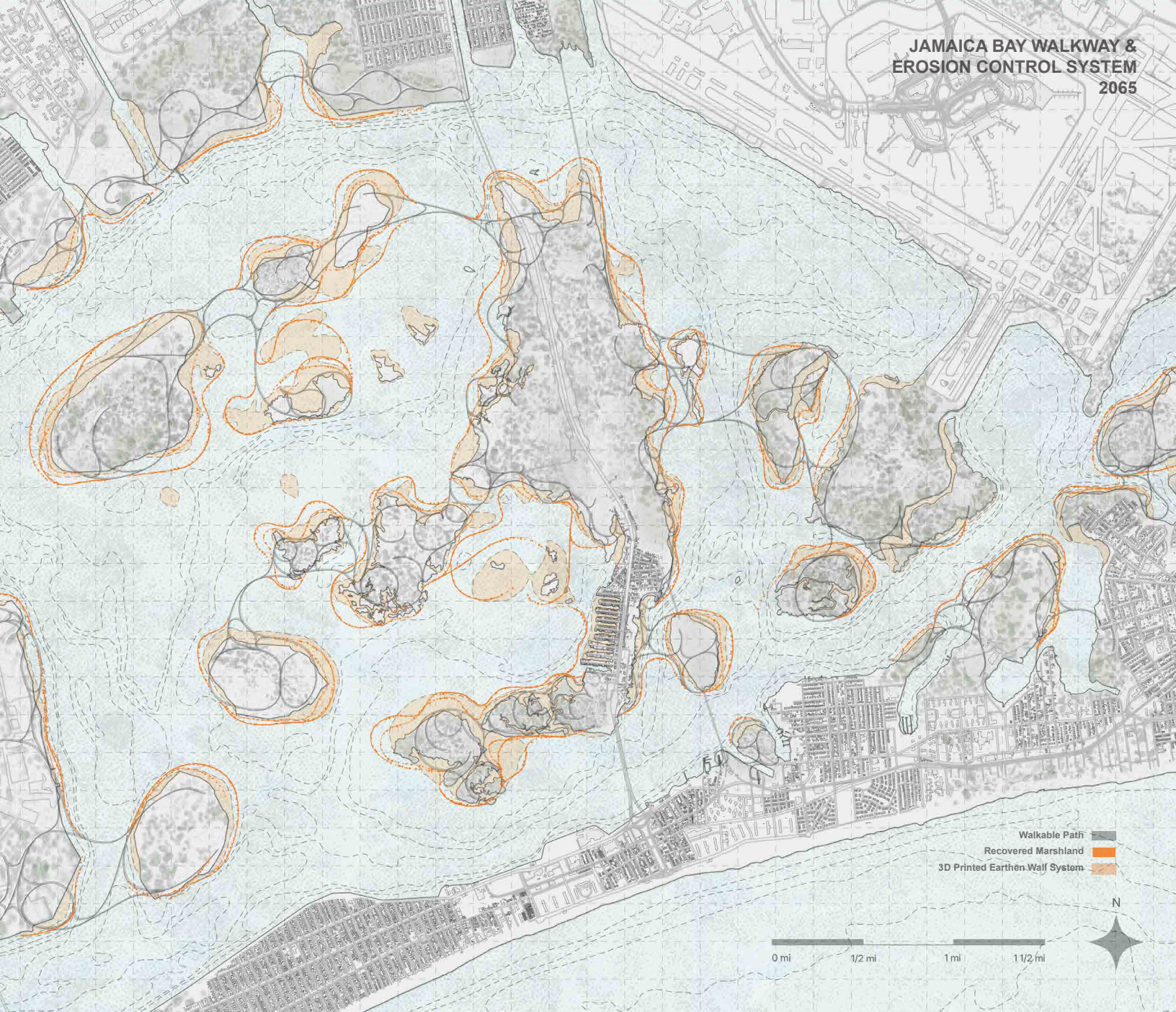
Section A - A'

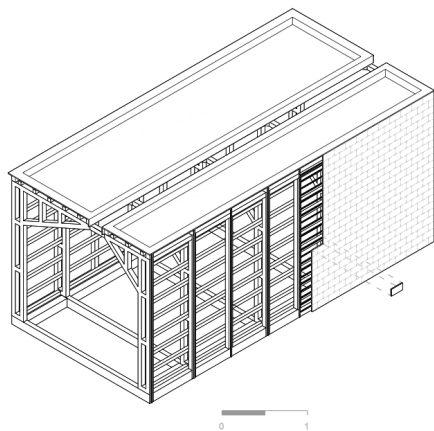


Section B - B'

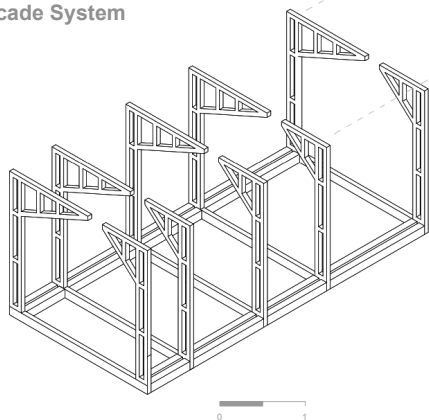




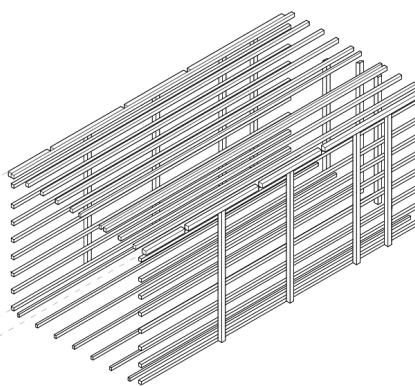




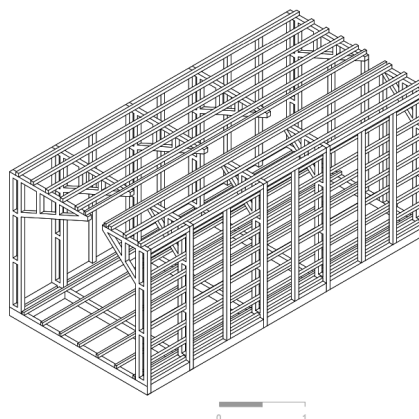
Structure - Facade System



Main Structure-Frames

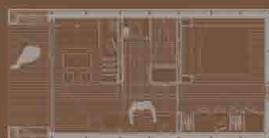
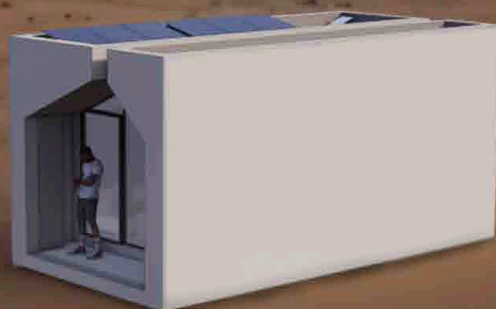


Total Structure Design



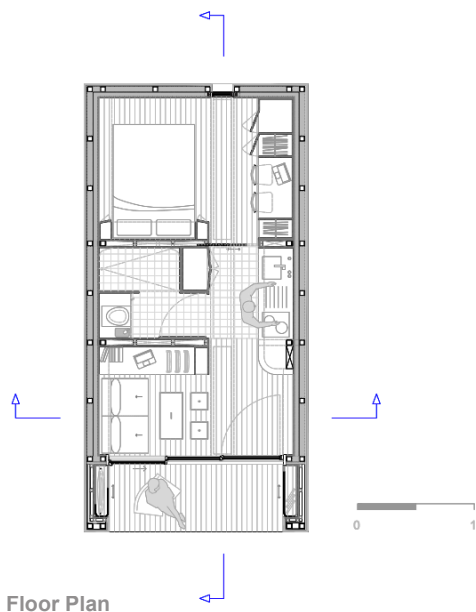
Render/Photomontage: Self-sufficient module in arid desert, with occupant about to go for a run

MICRO-HOME COMPETITION





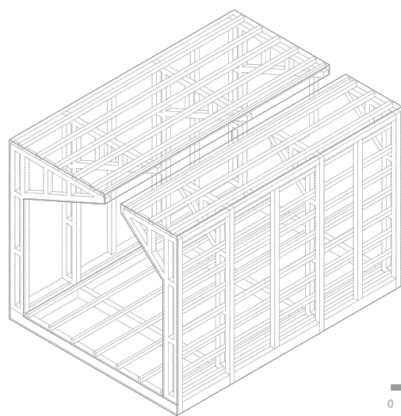
Render/Photomontage: Community concept of modules gathered around a point in an arid and desolate place



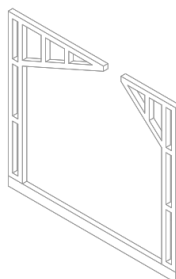
Open Competition.

Main task was to develop a 25sqm modular home which could be placed in an arid place for an extended period of time, requiring it to practice self-sufficient systems which would allow the design to stay off-grid for that long.

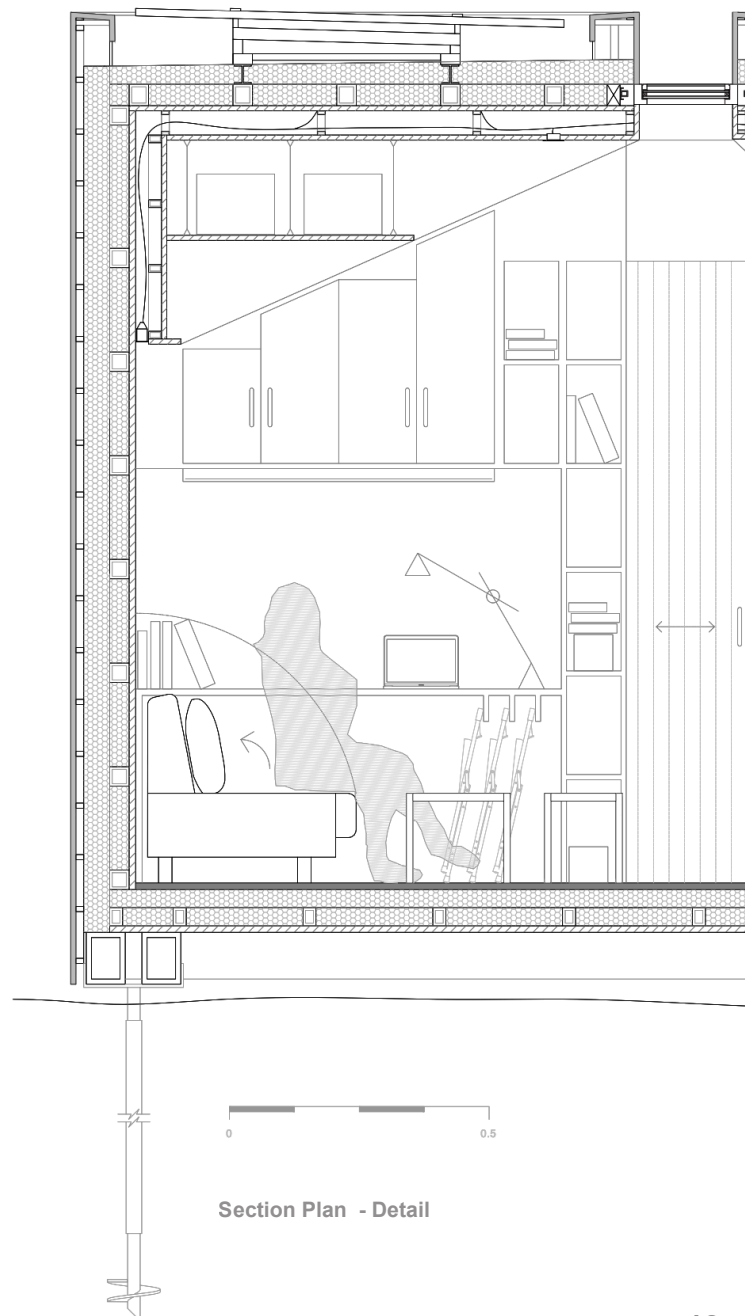
In this case, the light structure is covered by slick ceramic panels. Energy is stored in batteries on the roof, right next to the solar panels. A rain-water collector allows it to extend its resources beyond what the integrated water tank may provide.

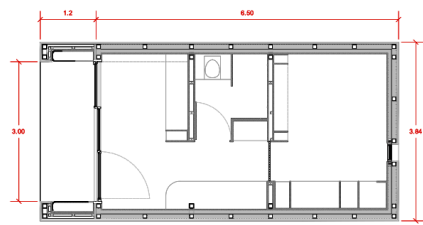


Axonometric - Empty Module

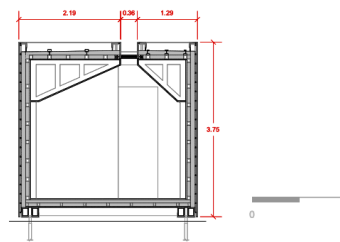


Frame Module

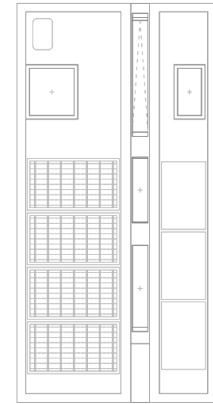




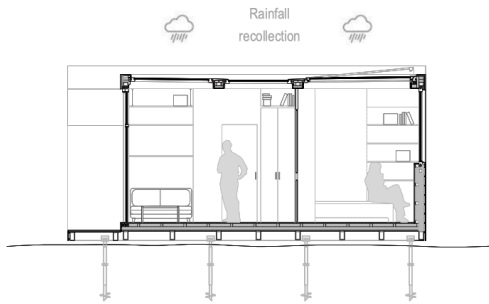
Empty Plan



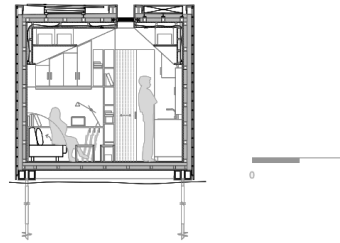
Empty Section



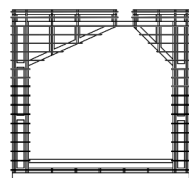
Ceiling Plan



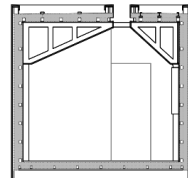
Empty Plan



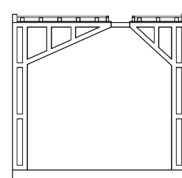
Empty Section



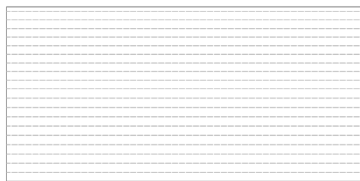
Facade Structure



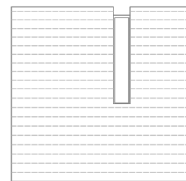
Construction Detail



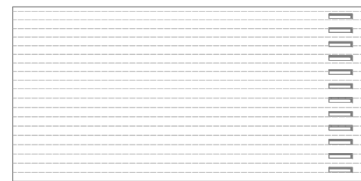
Structure



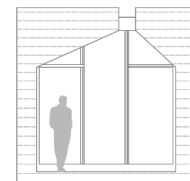
East Elevation



North Elevation



West Elevation



South Elevation

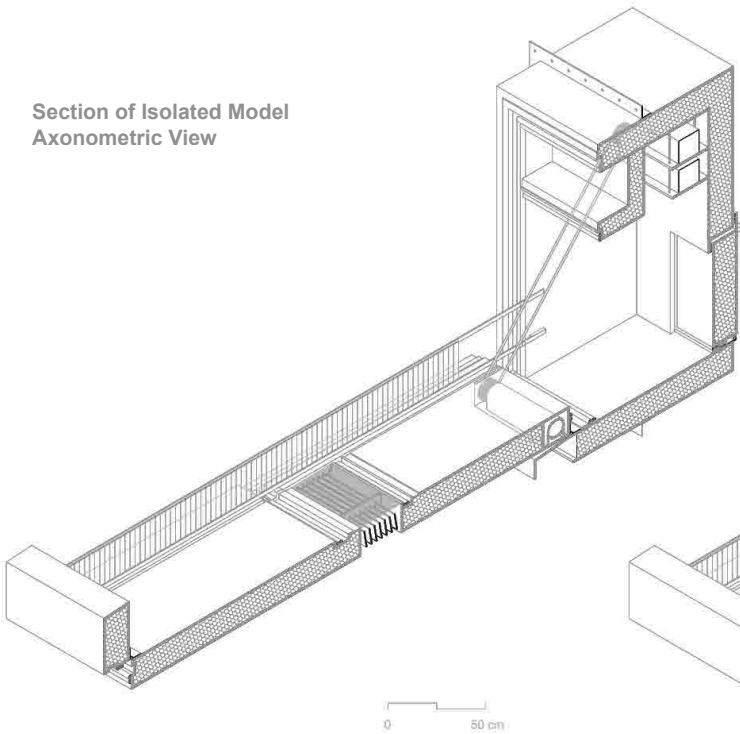
Render/Photomontage: Placement of module in an urban community, showcasing the versatility of the design



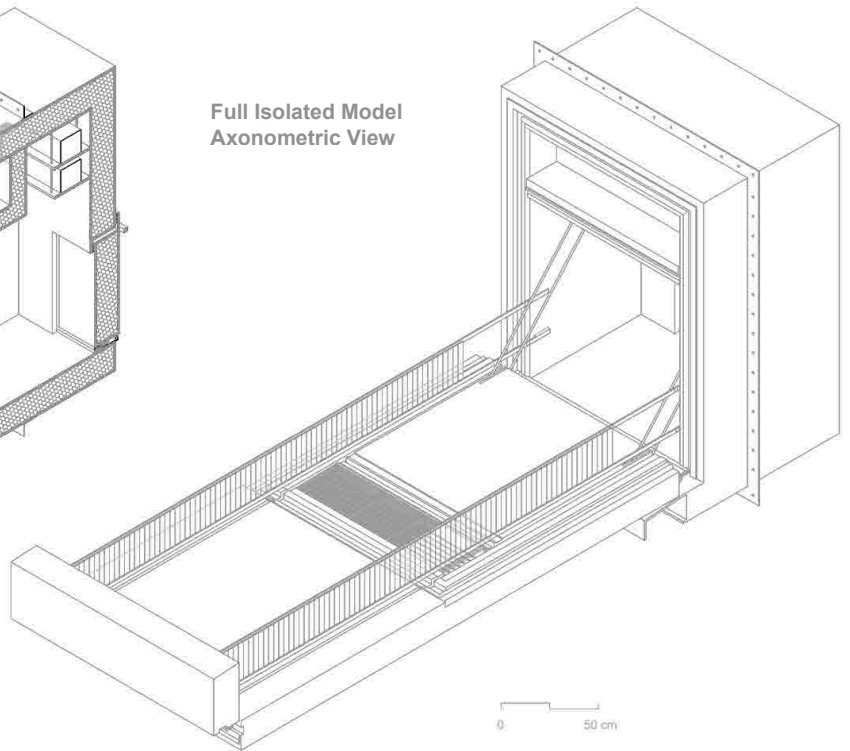
UAV FACADE DELIVERY SYSTEM B.ARCH PROJECT

Project completed during my B.Arch's last year.
Later published with faculty in a research paper on UAV Delivery Systems within dense urban areas, where delivery systems can be enhanced through the logistics medium provided by technological developments in drone architecture as well as the availability of unused vertical space in cities.

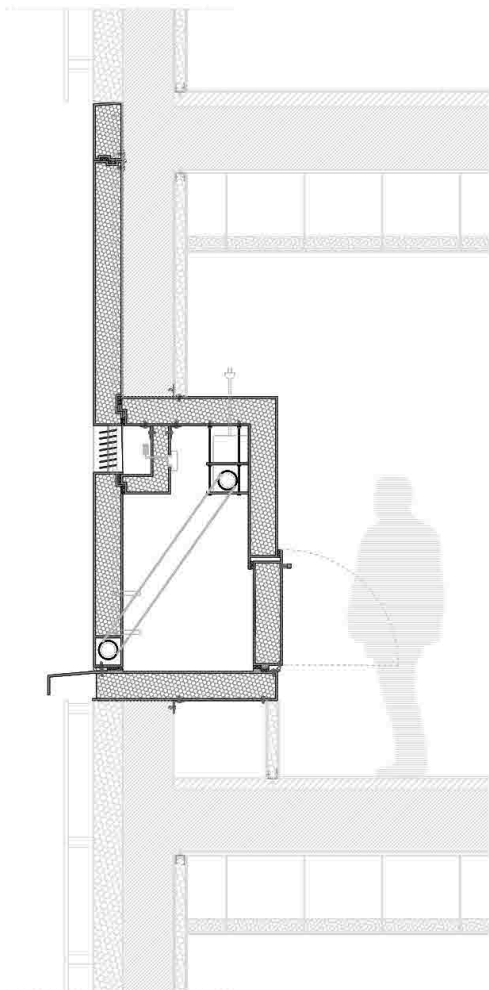
Section of Isolated Model
Axonometric View



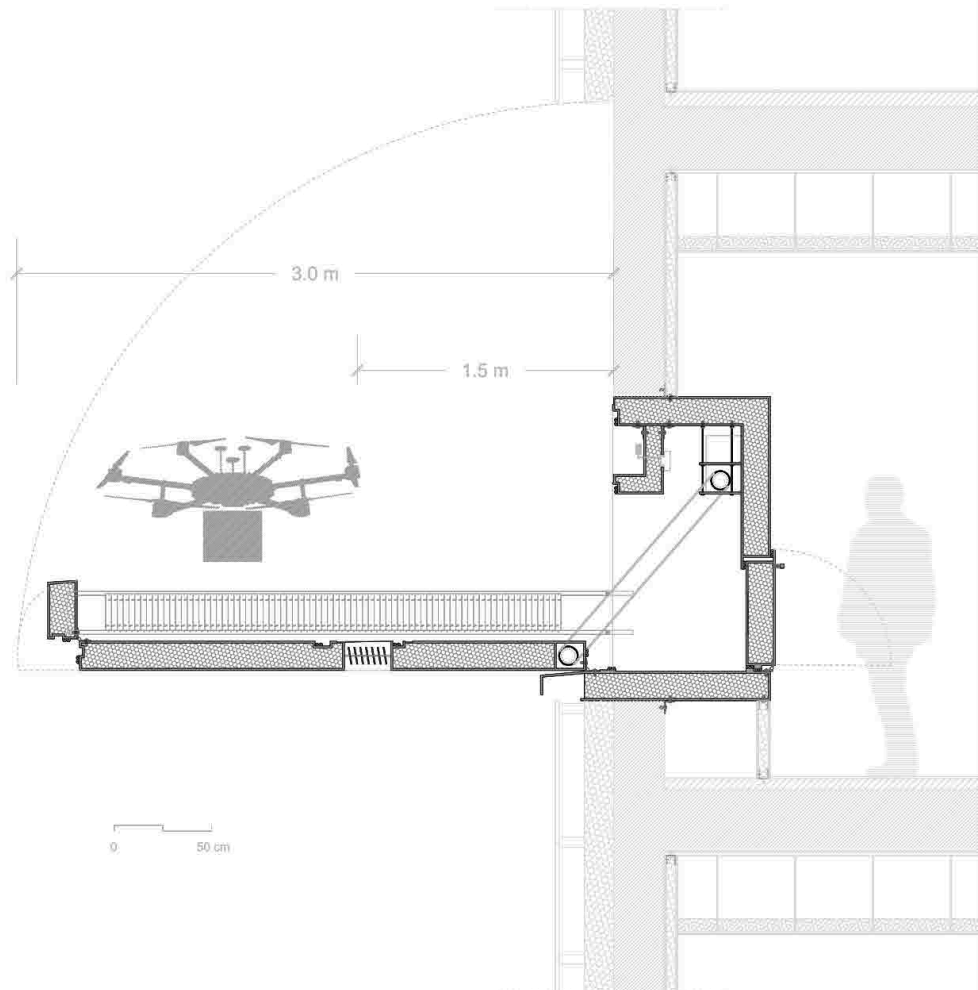
Full Isolated Model
Axonometric View



Section of Placed Model - Closed

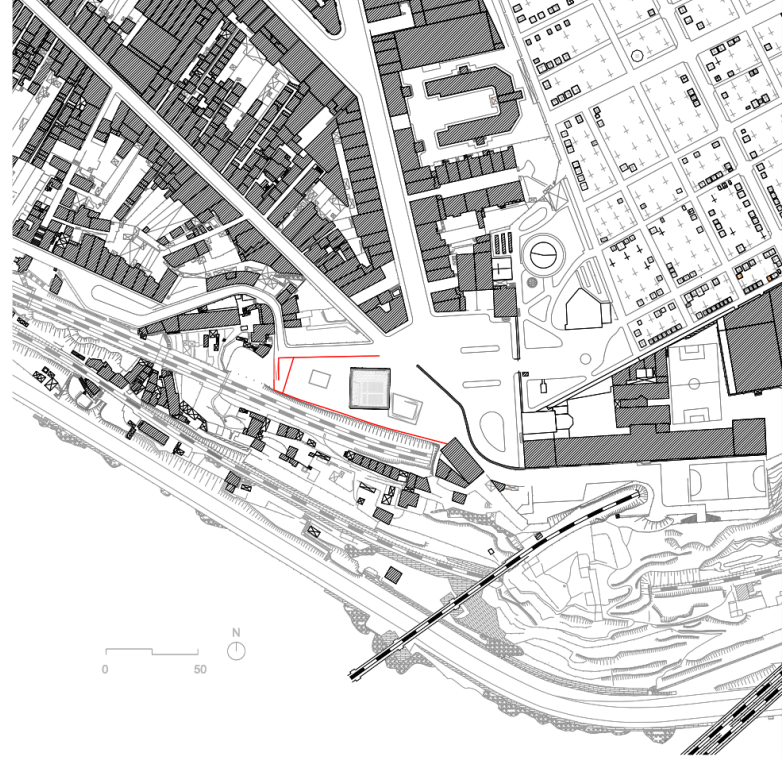
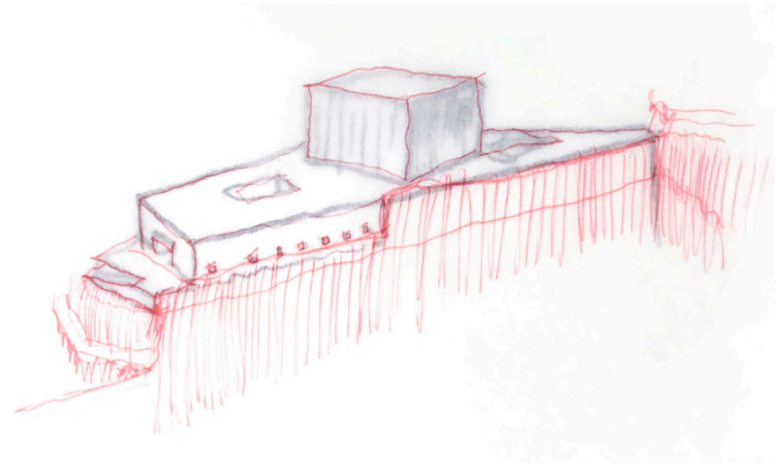


Section of Placed Model - Open

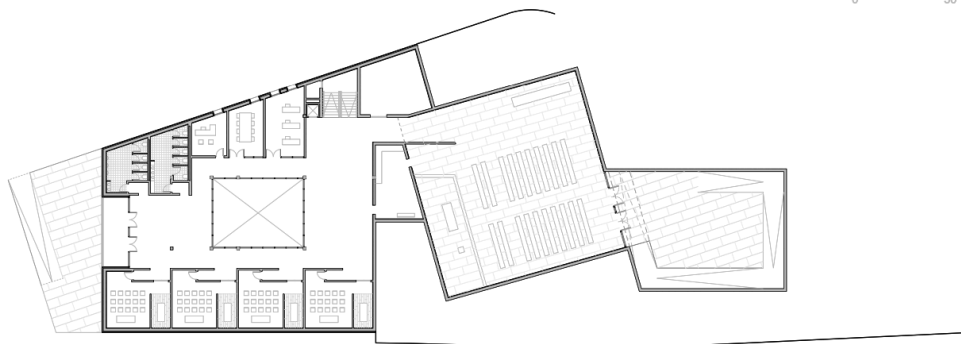


CHURCH IN PORTO

B.ARCH DESIGN STUDIO



Location Plan

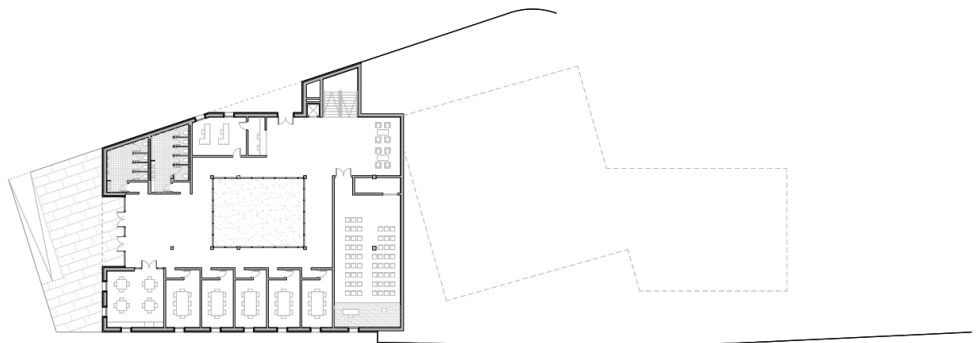


Under Ground Floor Plan

Project for the last year of my B.Arch.

Church and Parish complex, in the outskirts of Porto, overlooking the Douro River.

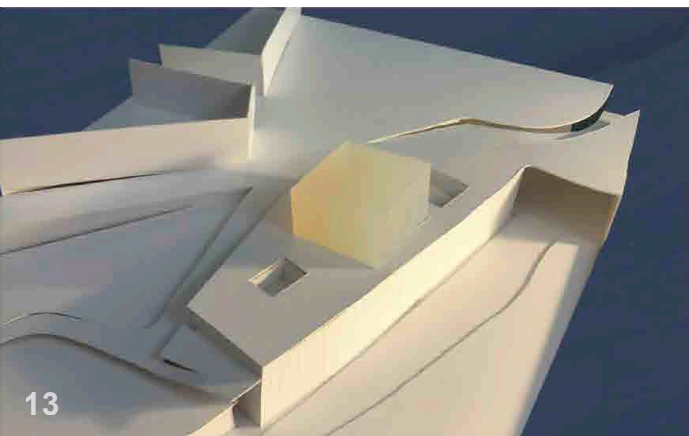
Structurally based design, with a solid base that creates a solid ground at a steep ledge where the rest of the program is placed, mainly in the inside, with the Church itself protruding out vertically, in a symbolic religious gesture.



Ground Floor Plan

Aerial View of White Model with Glicerine Church Element

Concrete + Glicerine Model Placed on Carboard Site Model

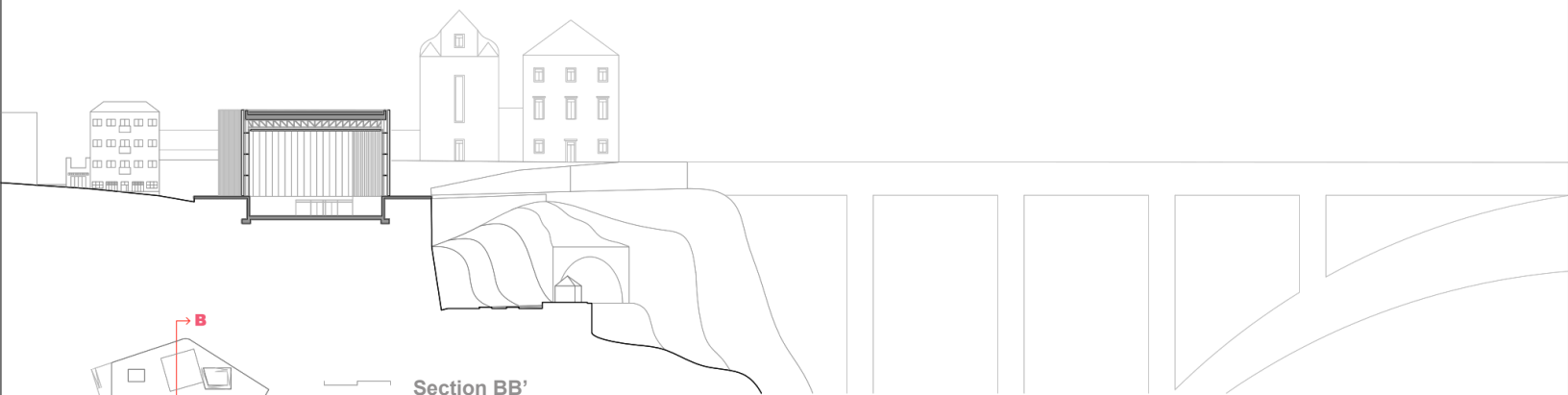
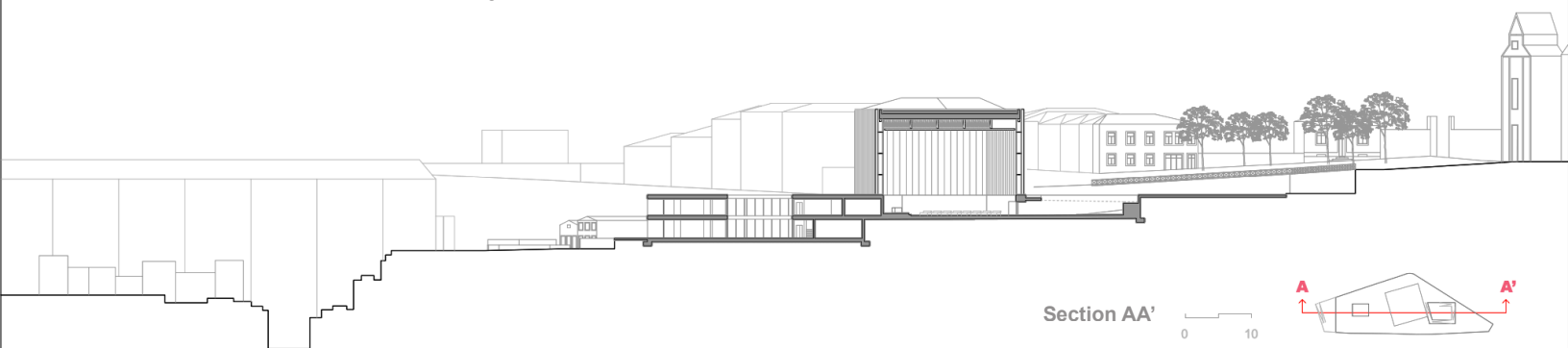


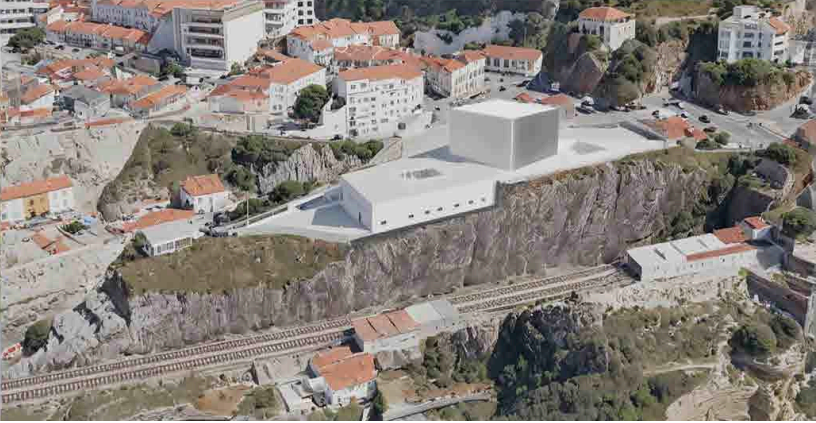


Render: Aerial view of the Model overlooking the Duoro River

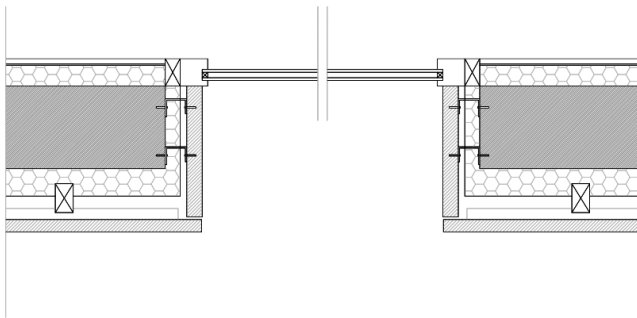


Interior Space of the Church

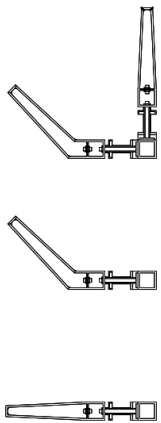




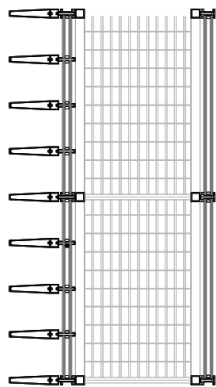
(Left) Eye-sight view from the Generated Plaza Space on Top (Right) Aerial View of Design on Ledge



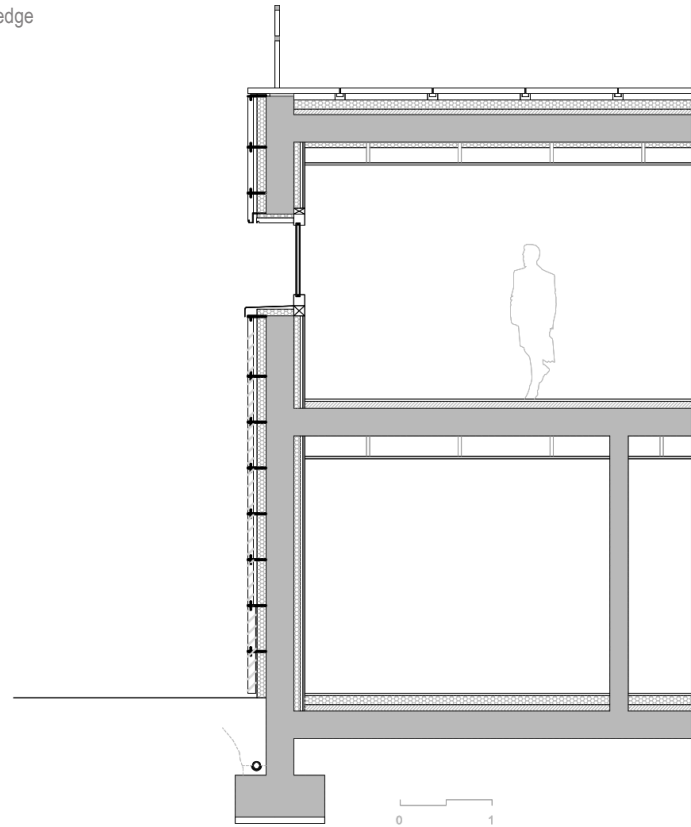
0 0.25 **Window Detail**



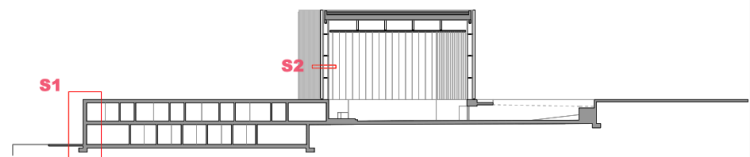
0 0.10 **Vertical Facade Elements**



0 0.25 **Facade Detail S2**



Facade Section S1

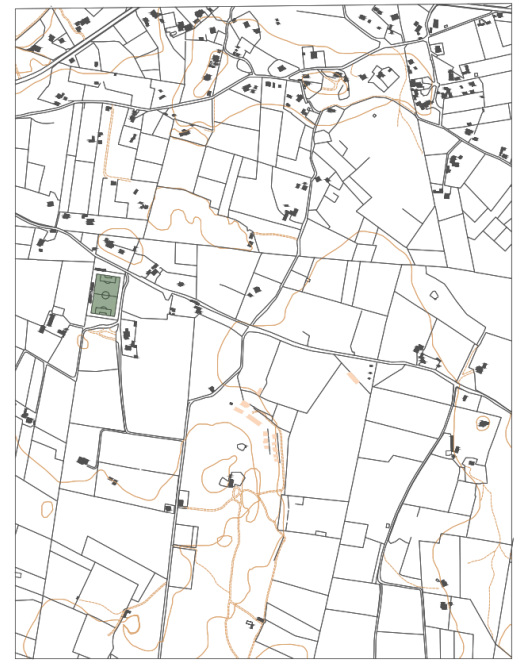


South-side Elevation

HOTEL BOUTIQUE

B.ARCH DESIGN STUDIO

Last project for my B.Arch. Design for an Hotel Boutique in the island of Formentera, Spain. The hotel complex is designed around the rising agricultural tourism in the island, where tourists can visit different plantations and even engage in procure or product tasting.

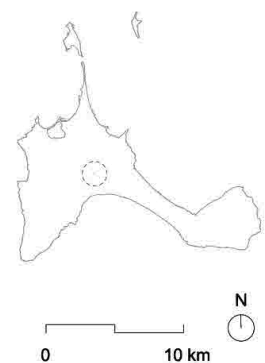


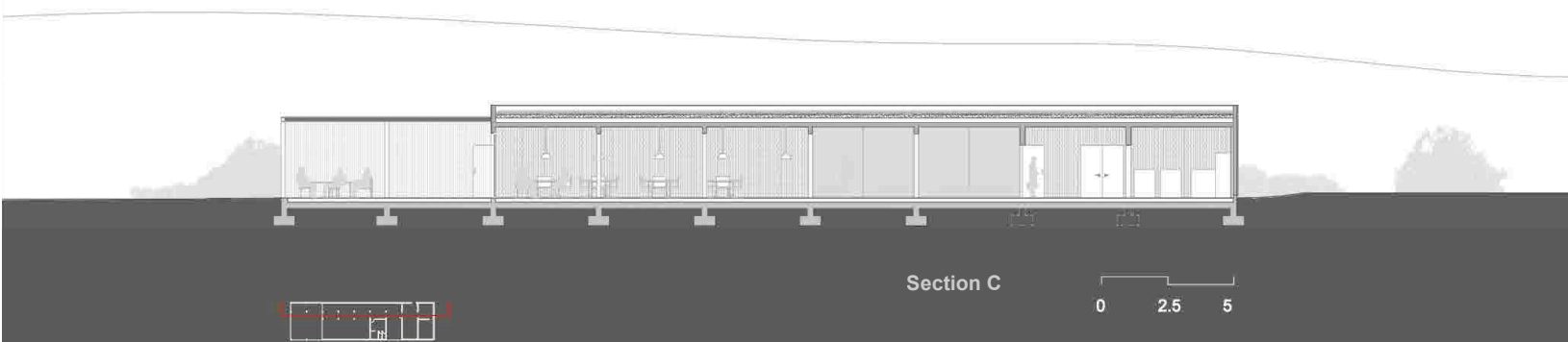
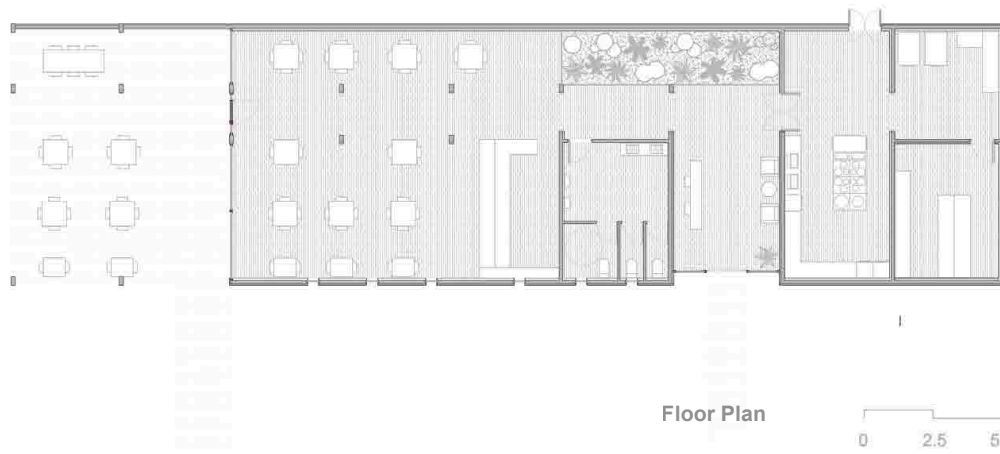
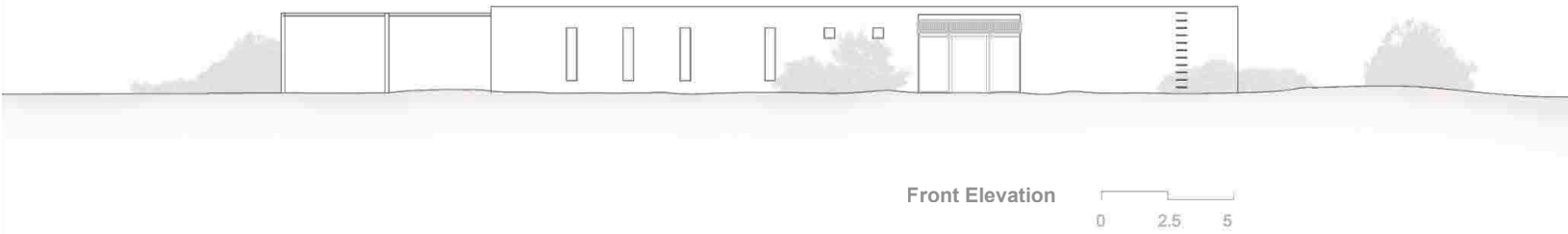
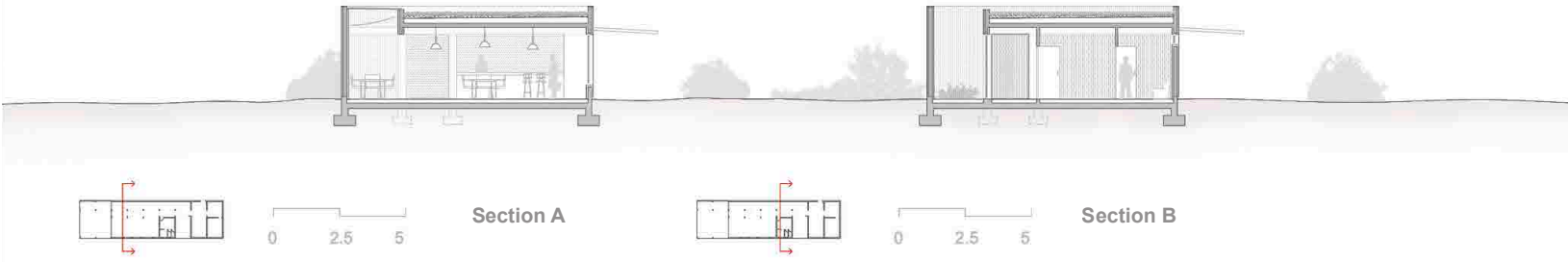
Plot Plan

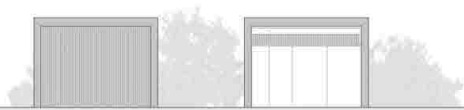
A total of 10 rooms, 2 suites, and 8 small bedrooms are placed along the tree fronts within the site. The preexisting building is refurbished to house the lounge as well as the lobby.

An underground spa and restaurant further enhance the Hotel, creating comfortable spaces where guests can rest after a long sunny day at the beach.

Formentera Island



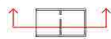




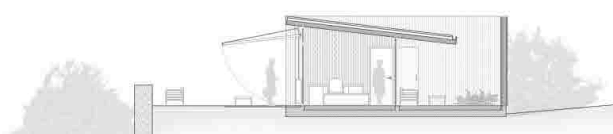
Small Rooms Elevation



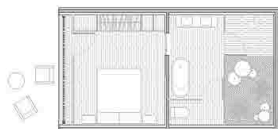
Suite Elevation



Small Rooms Section



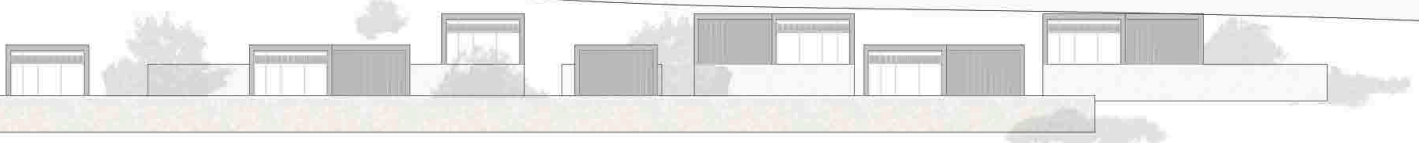
Suite Section



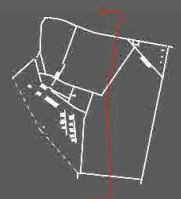
Small Rooms Floor Plan



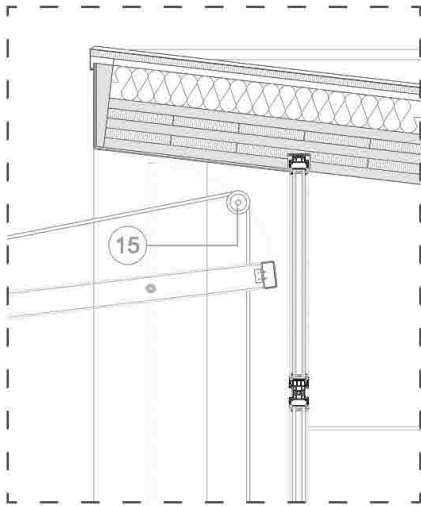
Suite Floor Plan



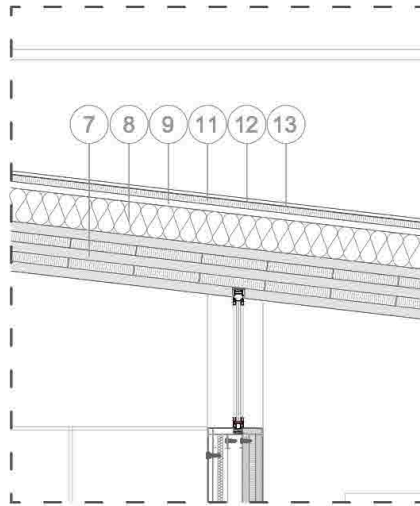
East Elevation



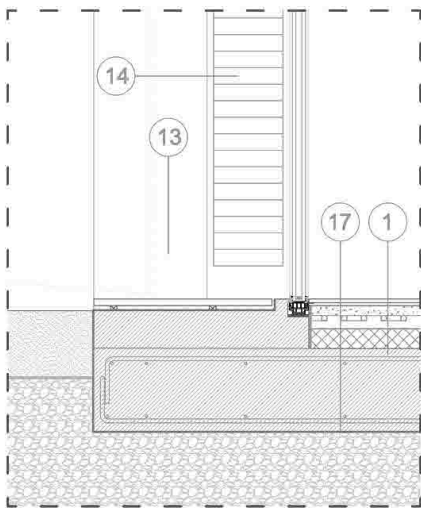
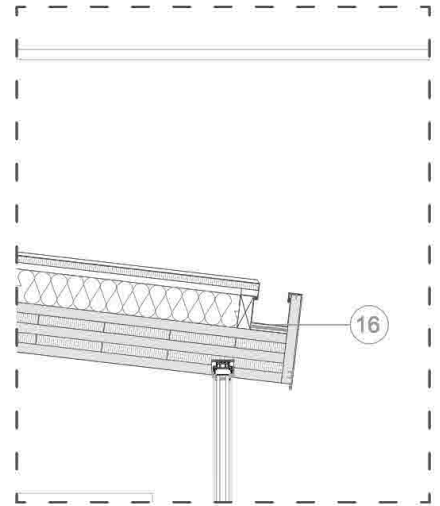
D01



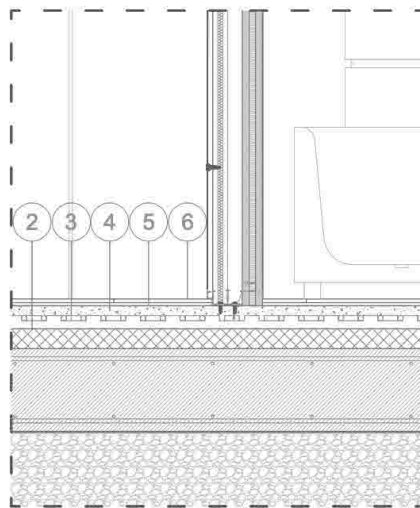
D02



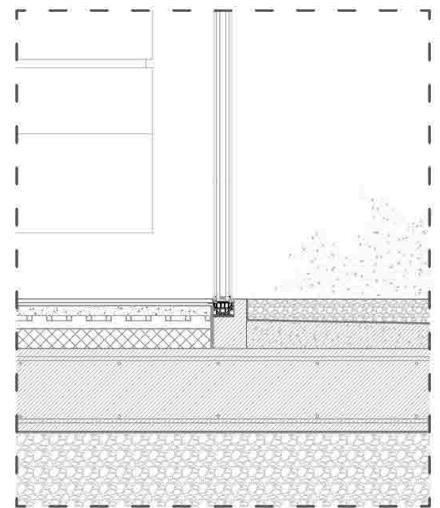
D03



D06



D05



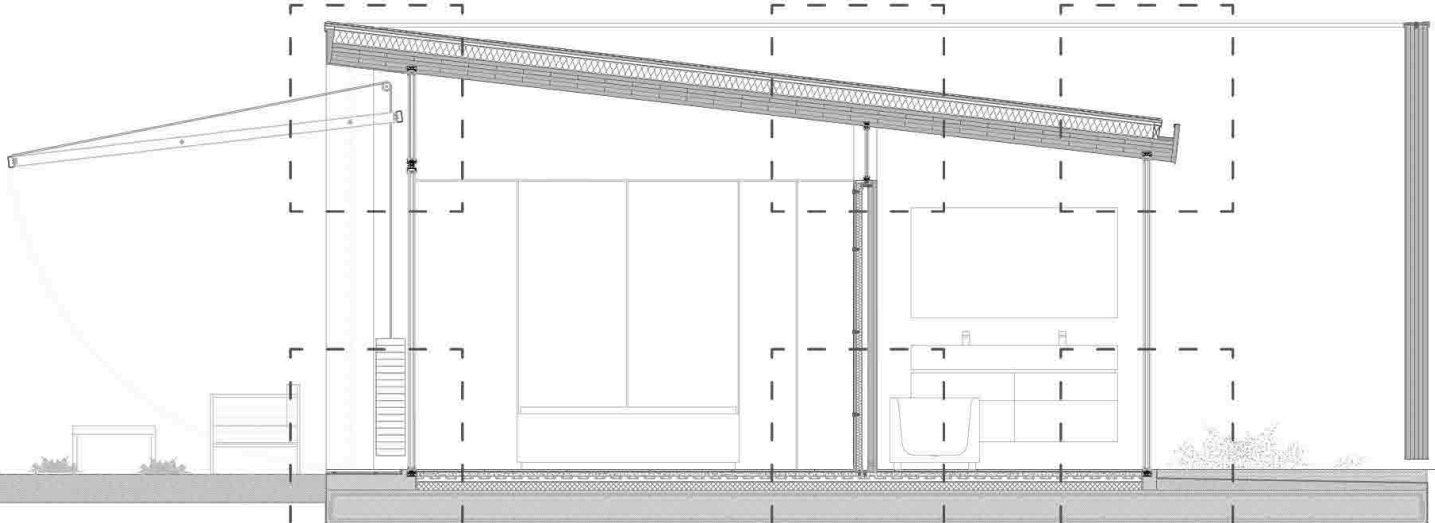
D04

0 0.5

D01

D02

D03



D06

D05

D04

0 0.5 1 Room Section Details

